

Sidhant Maloo

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EDUCATION

Huntley High School, Huntley, IL — Junior

Expected May 2028

GPA 3.97 unweighted / 4.62 weighted

AP Exams Computer Science A (5), Computer Science Principles (5), Precalculus (5), Human Geography (5), U.S. Government & Politics (5), Physics 1 (4)

2026–27 Courses AP Calculus BC, Statistics, Economics, English Language; PLTW Digital Electronics; Honors Chemistry

Independent Study MIT OpenCourseWare: 18.01 Single Variable Calculus (done), 18.06 Linear Algebra (in progress)

SKILLS

Programming Python, Java, JavaScript, HTML/CSS, Qiskit; machine learning (SVR, Random Forest, neural nets)

Hardware & Robotics VEX V5, mechanical design, autonomous systems, PCB design (KiCad), circuit simulation (Multisim), soldering, Raspberry Pi; GPS, vision, bumper, and color sensors

Leadership Team leadership, public speaking, mentoring, event organization

Languages English (fluent), Hindi (fluent), Spanish (conversational)

PROJECTS & RESEARCH

SVR vs. Random Forest on Volatile, Noisy Stock Data (Python)

Completed 2026

- Built a machine learning model that scores a stock's potential from multiple criteria, then wrote a research paper comparing how Support Vector Regression and Random Forest models hold up as noise and volatility increase.
- Paper is publicly posted on Zenodo (zenodo.org/records/20818947); code on GitHub. Actively seeking expert feedback.

Humanoid Robot ("Bob") (Python)

May 2026 – Present

- Building an AI-controlled upper torso with two 3-DOF arms and a head with a camera, microphone, and speaker to respond to natural spoken requests. Software is largely complete; the physical body is next.

Interactive Simulation Labs (six web-based simulations)

Completed 2026

- **Qubit Lab:** 1–3 qubit state engine with a Bloch sphere, H/X/Z/S, CNOT and Toffoli gates; entanglement computed from the reduced density matrix.
- **Event Horizon Lab:** real-time N-body black hole sandbox with gravitational-wave orbital decay and raymarched light bending.
- **IK Lab:** compares four inverse-kinematics solvers (FABRIK, CCD, Jacobian Transpose, Jacobian DLS) with live error tracking.
- **Synapse Lab** (backprop-trained digit recognizer), **Pathfinder Lab** (six search algorithms incl. BFS, Dijkstra, A*), and **Circuit Lab** (series/parallel circuit physics).

Hardware & Electronics

2020 – 2026

- Designing a custom PCB in KiCad (in progress, started May 2026); built a Raspberry Pi cyberdeck and used it to teach computing and quantum basics to students at Leggee Elementary.
- Built a soldering-station fume extractor and touch-controlled light (TTP223 sensors); analyzed circuits in Multisim; diagnosed and repaired a Sky Viper drone.

Software Projects

- **Java Minesweeper:** complete object-oriented game; planning STEM-education features and iOS/Android release.
- **Qiskit demonstration:** interactive quantum computing intro built for the Leggee Elementary Coding Club.
- **Custom boot dashboard:** live system stats, weather, and quote of the day.

EXPERIENCE & LEADERSHIP

HHS Robotics — Effort to Save the Program 2026 – Present

- After the club lost its head coach and funding for the 2026–27 school year, sent dozens of emails to administrators and met with the school board to request a coach and support.
- Still working to secure a coach and funding so the program can continue.

VEX V5 Robotics State Team — Builder, Designer & Programmer Aug 2024 – May 2026

- Programmed autonomous and driver-controlled behavior using GPS, vision, bumper, and color sensor logic.
- Contributed to mechanical design and iterative prototyping that improved robot performance.
- Helped earn the **Think Award** for documented engineering design and analysis.

Code Ninjas, Algonquin, IL — Sensei, Upper Belt Lead Apr 2026 – Present

- Supervise and tutor students across a wide age range in software development, including Unity and Java.

State Math Team (ICTM) — Captain Aug 2024 – Present

- Lead and organize practices for the 6-student state team; mentor teammates on problem-solving strategy and competition readiness.
- **2× regional champion** and state competitor; multiple medals. Recognized at HHS Breakfast of Champions.

North Central District Student Council — Treasurer Aug 2024 – Present

- Lead and organize large groups of students from schools across Illinois in the "student council of student councils."

Huntley High School Student Leadership Aug 2024 – Present

- **Principal Student Leadership Council** (Student Rep., Jan 2025–Present): represent the student body on school policy and initiatives affecting hundreds of students.
- **Athletics & Activities Leadership Team** (Student Rep.): advocate for athletics and extracurricular programs with administration.
- **Student Council** (Volunteer): plan and lead school-wide events attended by hundreds of students.
- **HHS Welcome Party** (Event Leader, Jan 2025–Present): organize one-on-one and small-group support for new and transfer students.

COMMUNITY SERVICE

Community Volunteer, Huntley, IL Oct 2022 – Present

- Completed **500+ hours** of service supporting the Huntley library and Farmers' Market.
- Started and mentor community **chess and coding clubs** for elementary students at local schools and libraries.

AWARDS & RECOGNITION

UN Ambassador Award — Inclusive outreach programs on the future of STEM 2026

Illinois House of Representatives Honors — State-recognized math skills 2025

Perfect PreACT 9 and PreACT 10 Scores — Standard college readiness benchmark 2025

Illinois Principals Association Student of the Year — Academic excellence 2024

INTERESTS

Music (flute, piano, harmonica, Native American flute, bugle, recorder, digital audio production), photography, Formula 1, philosophy, and reading.